

Utku Okur

Contact information

Faculty of Mathematics and Computer Science, TU Bergakademie Freiberg

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Born: January 5, 1992 — Izmir, Turkey

Current position

2026– Postdoc, Institute of Discrete Mathematics and Algebra, TU Bergakademie Freiberg, Germany

Areas of interest

Discrete mathematics, algebraic graph theory, combinatorics, formal verification (Lean 4)

Education

- 2020–2025 PhD in Mathematics, University of South Carolina, USA. Advisor: Joshua Cooper.
Thesis: *Coefficients of the characteristic polynomial of a hypergraph and their combinatorial properties*
- 2018–2020 MSc in Mathematics, Bilkent University, Turkey. Advisor: Laurence Barker.
Thesis: *Mackey decomposition for Brauer pairs*
- 2013–2018 BSc in Mathematics and BA in Philosophy, Boğaziçi University, Turkey

Research publications

8. Aida Abiad, Joshua Cooper, Utku Okur, *A general switching method for constructing E-cospectral hypergraphs*, Preprint (2026). [arXiv:2604.06016](https://arxiv.org/abs/2604.06016)
7. Ann Clifton, Éva Czabarka, Audace A. V. Dossou-Olory, Kevin Liu, Sarah Loeb, Utku Okur, László Székely, Kristina Wicke, *Reconstruction of Planar Tanglegrams*, In preparation (25+).
6. Ann Clifton, Éva Czabarka, Kevin Liu, Sarah Loeb, Utku Okur, László Székely, Kristina Wicke, *Reconstruction of caterpillar tanglegrams*, Discrete Applied Mathematics 378 (2026), 647–661. [arXiv:2501.13670](https://arxiv.org/abs/2501.13670) doi:10.1016/j.dam.2025.09.022
5. J. Cooper, U. Okur, *Multigraphs with Unique Partition into Cycles*, Submitted (25+) to Discussiones Mathematicae Graph Theory. [arXiv:2504.08083](https://arxiv.org/abs/2504.08083) doi:10.7151/dmgt.2627
4. J. Cooper, U. Okur, *Partition Lattice of an Eulerian Digraph*, Submitted (25+) to Discrete Mathematics. [arXiv:2502.00867](https://arxiv.org/abs/2502.00867)
3. Ann Clifton, Éva Czabarka, Audace A. V. Dossou-Olory, Kevin Liu, Sarah Loeb, Utku Okur, László Székely, Kristina Wicke, *Decks of rooted binary trees*, European Journal of Combinatorics 124 (2025), 104076. [arXiv:2311.02255](https://arxiv.org/abs/2311.02255) doi:10.1016/j.ejc.2024.104076

2. J. Cooper, K. Guo, U. Okur, *Characteristic Polynomials and Hypergraph Generating Functions via Heaps of Pieces*, Submitted (26+) to Linear and Multilinear Algebra. [arXiv:2411.03567](https://arxiv.org/abs/2411.03567)
1. J. Cooper, U. Okur, *Polynomial reconstruction problem for hypergraphs*, Linear Algebra and its Applications 692 (2024), 37–61. [arXiv:2312.16152](https://arxiv.org/abs/2312.16152) doi:10.1016/j.laa.2024.03.029

GitHub repositories

- Hypergraph calculations:
<https://github.com/utkuokur/Hypergraph-Programs>
- Tanglegram calculations:
https://github.com/utkuokur/binary_trees_and_tanglegrams

Other writings

- Vaibhav Bajpai, Nathan Bowler, Johannes Carmesin, Utku Okur, *10 Scalable Challenges in Combinatorics: A Public Benchmark Formalised in Lean*, Extended abstract, AI4Math workshop at CICM 2026, Ljubljana; informal proceedings, [PDF](#).
- Vaibhav Bajpai, Johannes Carmesin, Utku Okur, *A Lean 4 Formalisation of OpenAI's Proof of the Cycle Double Cover Conjecture*, Lean code: [GitHub](#), doi:10.5281/zenodo.21410822. See also J. Carmesin, *The cycle double cover theorem*, The Matroid Union, July 2026.
- Ann Clifton, Éva Czabarka, Kevin Liu, Sarah Loeb, Utku Okur, László Székely, Kristina Wicke, *Universal rooted phylogenetic tree shapes and universal tanglegrams*, arXiv preprint (2023), [arXiv:2308.06580](https://arxiv.org/abs/2308.06580); not intended for publication, since a better result exists; see Gawrychowski, Kuhn, Łopuszański, Panagiotou, Su, SODA 2018, [arXiv:1707.06011](https://arxiv.org/abs/1707.06011), doi:10.1137/1.9781611975031.166.

Talks

2024	Discrete Math and Combinatorics Seminar, University of South Carolina, Columbia/SC.
2023	<i>Invited Speaker</i> : Joint Mathematics Meetings, AMS Special Session on Trees in Many Contexts, Boston/MA.
2023	Graduate Mini-Conference in Discrete Mathematics, University of South Carolina, Columbia/SC.
2023	Discrete Math and Combinatorics Seminar, University of South Carolina, Columbia/SC.

Workshops attended

2026	Workshop on Artificial Intelligence for Mathematics (AI4Math) , Conference on Intelligent Computer Mathematics (CICM 2026), September 21, 2026, Ljubljana.
2022	Trees in Many Contexts, Mathematics Research Community (MRC) Conference, June 5–11, 2022, Boston.

Grants, scholarships & awards

2013–2018 TUBITAK Scholarship for Fundamental Sciences
2019 Orhan Alisbah Fellowship Award

Teaching summary

2018–2020 Teaching Assistant at Bilkent University, Turkey
2019–2020 Research Assistant at Middle Eastern Technical University (METU), Turkey
2020–2021 Teaching Assistant at the University of South Carolina (UofSC), USA
2021–present Instructor of Record at the University of South Carolina (UofSC), USA

Teaching experience

2020 *Teaching Assistant (TA)*: Math 142 – Calculus II
2021 *Instructor*: Math 115 (Precalculus Mathematics)
2022 *Instructor*: Math 115, Math 122 (Calculus for Business Administration & Social Sciences), Math 111 (Basic College Mathematics)
2023 *Instructor*: Math 115, Math 122, Math 241 (Vector Calculus)
2024 *Instructor*: Math 122, Math 111, Math 170 (Finite Mathematics)

Service

2024 *Organizer*: Discrete Math and Combinatorics Seminar, University of South Carolina
2023–2025 *Referee*: Linear and Multilinear Algebra, The Electronic Journal of Linear Algebra, Theoretical Computer Science, Discrete Mathematics